FMGT 3510 – Finance 1
Fall 2026

Chapter 4

Time Value of Money (TVM)

Overview of the Time Value of Money (1 of 3)

Example 1. Your friend **Webster owes you $100**. He is supposed to **pay you $100 back today**, but he would like to **pay you in 1 year instead**.

If you accept to **receive the $100 in one year** instead of today you are taking a **risk**:

- Webster may not be able or willing to pay you in 1 year (**default risk**)
- Due to **inflation**, $100 today has a higher **purchasing power** than $100 received in 1 year.

If you accept to **receive the $100 in one year** instead of today you are also making a **sacrifice**:

- You can NOT spend the $100 today, but will have to wait 1 year (delayed consumption)
- You can NOT invest the $100 today, you can not earn interest on this money for 1 year.

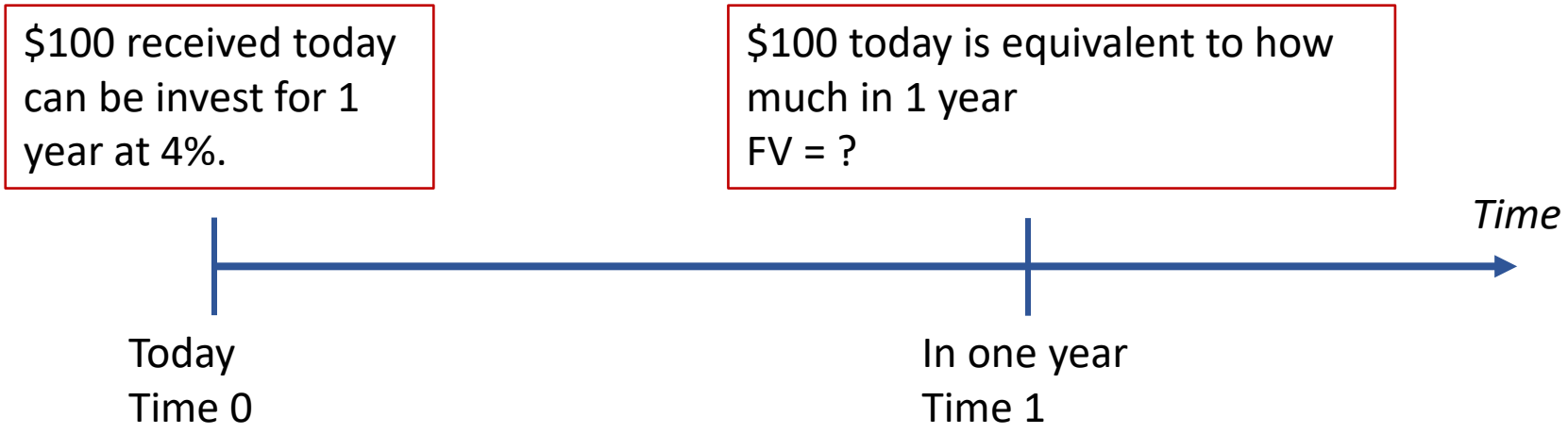
Conclusions

- **$100 today** (*present value*) is **worth more** to you **than $100 in one year** (*future value*). There is an **opportunity cost** associated with receiving the $100 later.
- You will accept Webster's offer to pay you in 1 year only if he agrees to pay you **$100 plus enough interest to compensate for this opportunity cost.**

Overview of the Time Value of Money (2 of 3)

Example 1. Continued

Assume that the prevailing relevant interest rate is 4% per year.



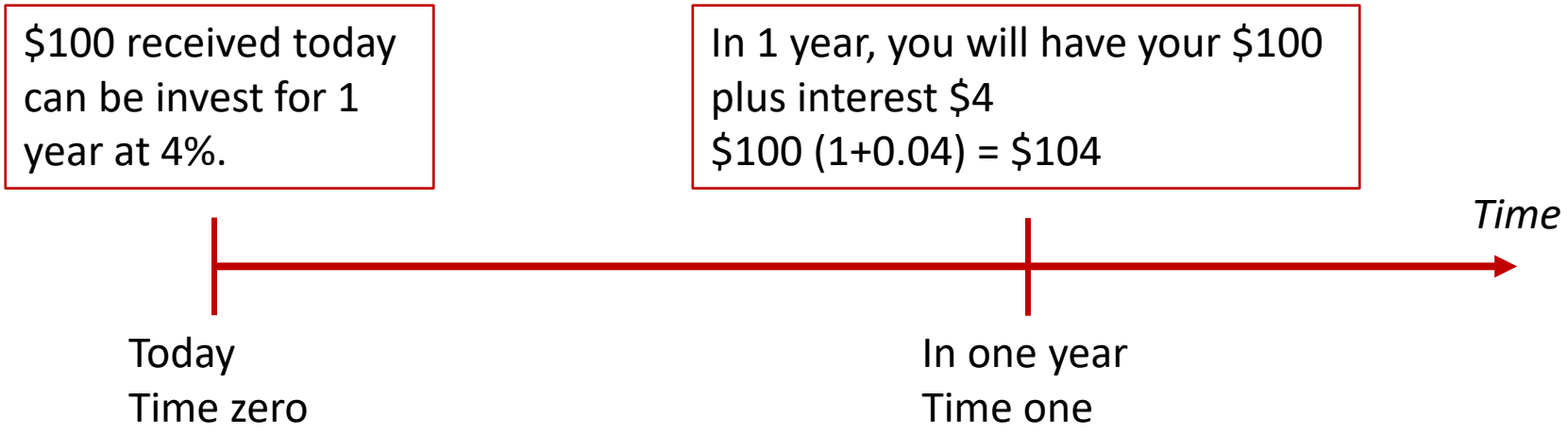
This first step in analyzing any financial project is to draw a **timeline**. The timeline allows us to visualize the cash flows and their timing. It reduces the risk of miscalculations.

- *Time 0 (t=0) corresponds to today.*
- *Time 1 means the end of the first period (end of the first year in our example), beginning of the second period.*
- *Cash outflows are shown as negative. Cash inflows are shown as positive.*

Overview of the Time Value of Money (3 of 3)

Example 1. End

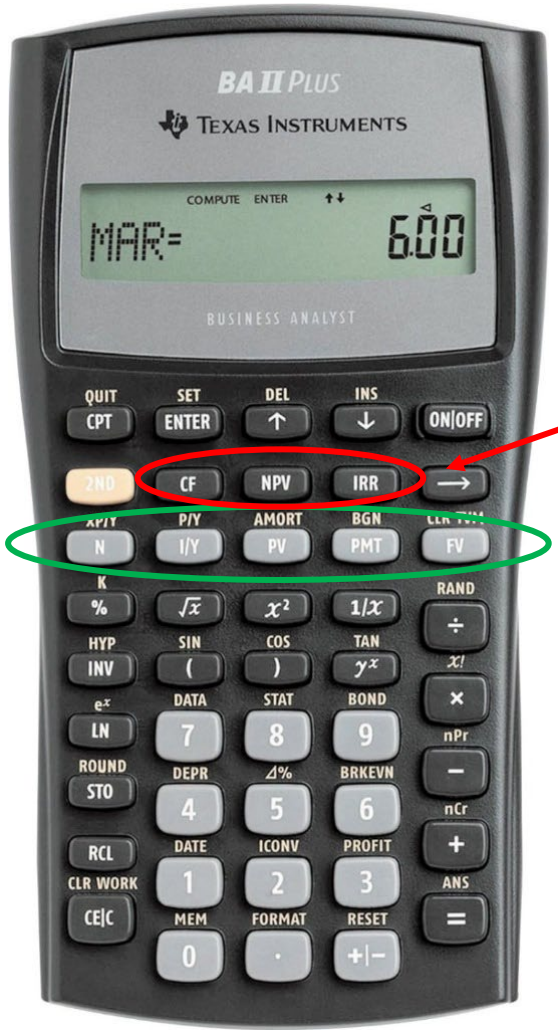
Assume that the prevailing relevant interest rate is 4% per year.



Conclusions

- **$100 today** (present value) **is worth the same** to you as **$104 in one year** (future value).
- The difference between the $100 (present value) and the future value of these $100, reflects the **time value of money**.

Financial calculator's guidelines (1 of 2)



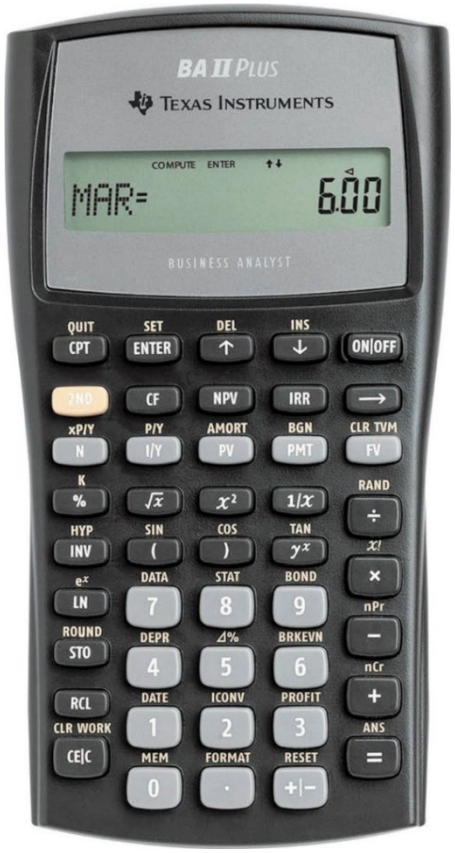
Time Value of Money (TMV) calculations can be done using your financial calculator.

They can also be done in Excel using its built-in financial functions. A list of these functions has been posted on Learning Hub.

CF ("cash flow") keys: for Net Present Value (NPV) of a series of **unequal** cash flows and for Internal Rate of Return (IRR) calculations.

Annuity keys: PV and FV of a single payment and for PV and FV of a series of **equal** cash flows.

Financial calculator's guidelines (2 of 2)



Most of the problems in this course **assume one payment per year**. You should therefore make sure that the payment frequency ("P/Y") and compounding frequency ("C/Y") registers are set to "1". We will address exceptions to this rule as required.

Always return your calculator to **"END" mode** if you used "BGN" mode to solve a problem with beginning of period payments.

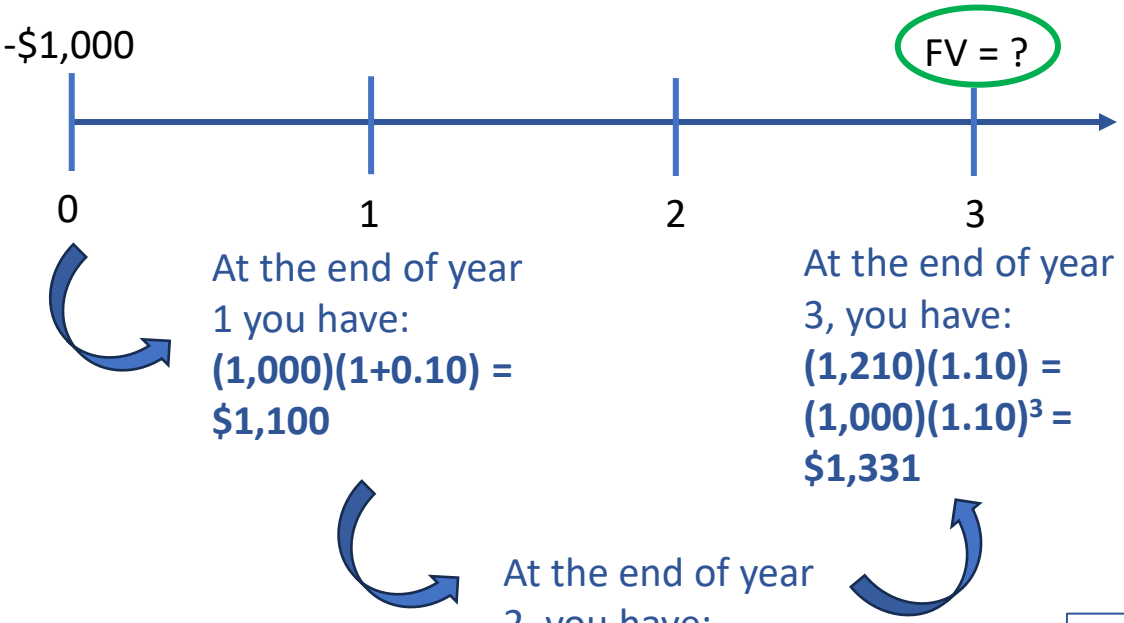
One of your cash flow inputs must be negative when using the annuity keys and when calculating IRR, otherwise you will receive an error message.

We generally calculate **interest rates** to **two decimal places e.g. 5.25%** or four places if the rate is expressed in decimal form e.g. 0.0525. **Do not provide answers that extend to some unreasonable number of decimal places.**

Remember to enter discount rates in percentage form, not decimal form, when using the TVM functions of your calculator.

Future value - lump sum – compound interest (1 of 3)

Example 2. You invest $1,000 today in a project that will earn 10% per year for three years. What will your investment be worth at the end of three years?



During years 2 and 3, we are earning interest on our original $1,000 (*principal*) as well as the **reinvested interest**.

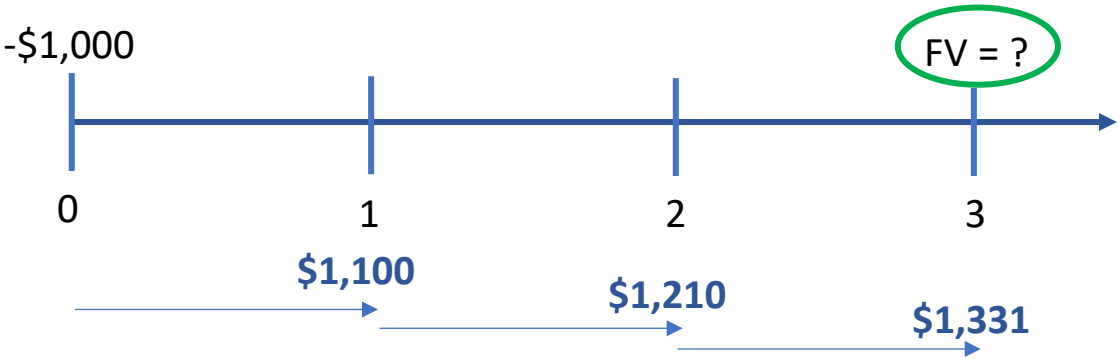
This "interest on interest" is called **compound interest**.

$$FV_N = PV_0(1 + I)^N$$

$$\mathbf{FV_3 = 1,000(1 + 0.10)^3 = 1,331}$$

Future value – lump sum – compound interest (2 of 3)

Example 2 end You invest $1,000 today in a project that will earn 10% per year for three years. What will your investment be worth at the end of three years?

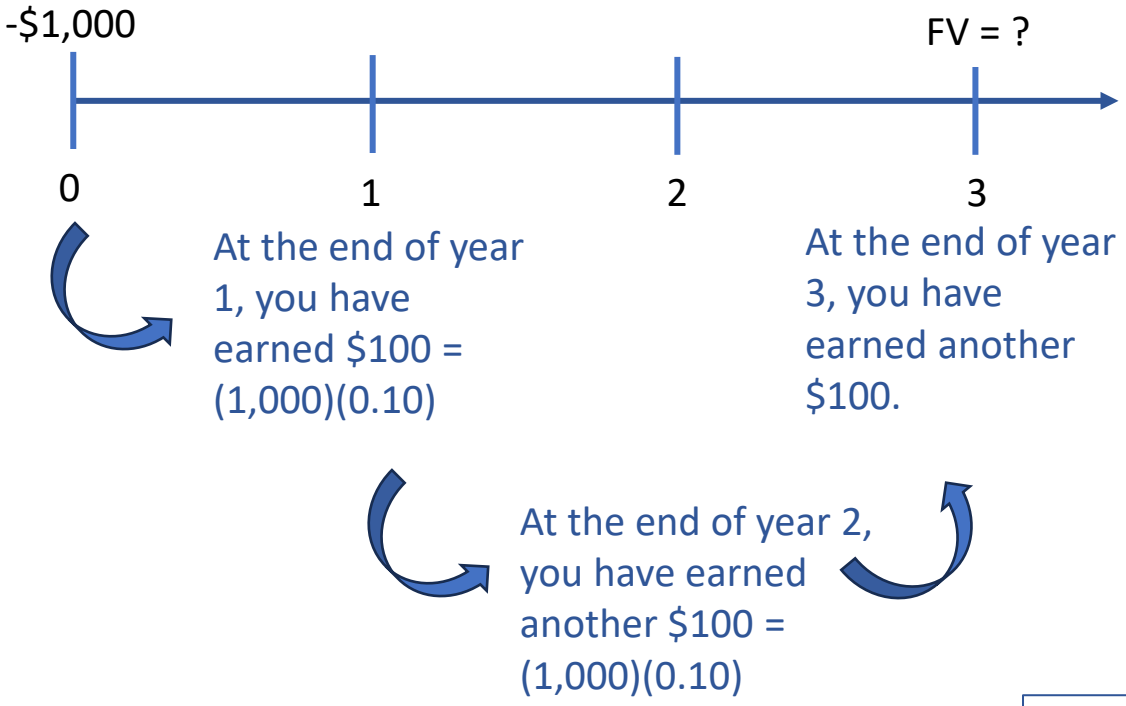


Using your financial calculator

	N	i/Y	PV	PMT	FV
at the end of year 1	1	10	-1,000	0	**1,100**
At the end of year 2	2	10	-1,000	0	**1,210**
At the end of year 3	3	10	-1,000	0	**1,331**

Future value – lump sum – simple interest (3 of 3)

Example 3. You invest $1,000 today in a project that will earn 10% per year for three years. What will your investment be worth at the end of three years?



Most TVM calculations assume that interest is reinvested and that additional interest is earned on interest received.

If we did not make that assumption, we end up with a **simple interest** calculation (i.e. interest earned only on the principal).

$$FV_N = PV_0(1 + (N)(r)$$

$$\mathbf{FV_3 = 1,000(1 + (3)(0.1) = 1,300}$$

Conclusion

FV in 3 years (compound interest) = $1,331

FV in 3 years (simple interest) = $1,300

Difference = $31 = interest earned on interest if compound interest.

Future value – simple vs. compound interest

Difference between simple and compound interest.

Assuming:

- Initial investment at tome 0 (today) is PV(0) = $1,000
- Interest rate per year is 10%

Let's calculate the investment future value (FV) in 5, 10, 15 and 20 years:

	FV(3)	FV(5)	FV(10)	FV(15)	FV(20)
Compound interest	$1,331	$1,610.5	$2,593.7	$4,177.3	$6,727.5
Simple interest	$1,300	$1,500	$2,000	$2,500	3,000
Difference *Interest earned on interest*	$31	$110.5	$593.7	$1,677.3	$3,727.5

Conclusions

- The **longer the investment period**, the **greater** the **difference** in FV.
- **Simple interest are normally used only for short periods of time** (up to 1 year).
- **Unless told otherwise, always assume compound interest.**

Future value – ORDINARY ANNUITY

Ordinary annuity: same amount, paid out (or received) once per period (ex. annually), always at the END of the period and for a limited time.

Example 4. You invest **$1,000** each year for the **next 3 years**, with the first $1,000 to be invested in 1 year. If the interest rate is 10% per year (annual rate of return), how much will you have in 3 years?



END	N	i/Y	PV	PMT	FV
Input	3	10	0	-1,000	
CPT					**3,310**

Make sure that your calculator is in END mode

$$\mathrm{FVA}_{\mathrm{ordinary}} = \frac{\mathrm{PMT}}{\mathrm{I}}[(1+\mathrm{I})^{\mathrm{N}} - 1]$$

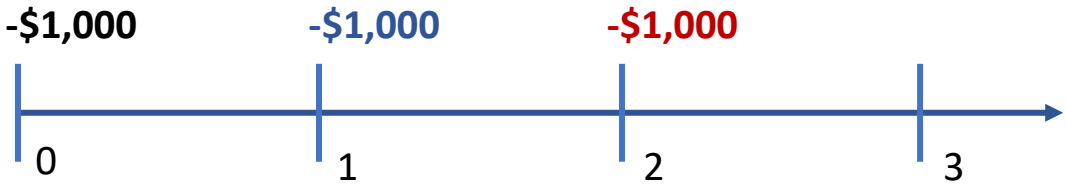
$$\mathbf{FVA}_{ordinary} = \left(\frac{\mathbf{1,000}}{\mathbf{0.10}}\right)((\mathbf{1+0.10})^{\mathbf{3}} - \mathbf{1}) = \mathbf{3,310}$$

$$\mathrm{FV}(2) = ((1{,}000)(1.10)) + 1{,}000 = \$2{,}100$$
$$\mathrm{FV}(3) = (\mathbf{1,000})(\mathbf{1+0.10})^{\mathbf{2}} + ((1{,}000)(1.10)) + \mathbf{1,000} = \$\mathbf{3,310}$$

Future value – ANNUITY DUE

Annuity DUE: same amount, paid out (or received) once per period (ex. annually), always at the **BEGINNING** of the period, for a limited time.

Example 5. You invest $1,000 each year for the next 3 years, with the first $1,000 to be invested today. If the interest rate is 10% per year (annual rate of return), how much will you have in 3 years?



$$\text{FVA}_{\text{due}} = \text{FVA}_{\text{ordinary}}(1 + \text{I})$$

$$\text{FVA}_{\text{due}} = \frac{\text{PMT}}{\text{I}}[(1 + \text{I})^{\text{N}} - 1](1 + \text{I})$$

$$\mathbf{FVA_{due} = 3,310(1 + 0.10) = 3,641}$$

BGN	N	i/Y	PV	PMT	FV
Input	3	10	0	-1,000	
CPT					**3,641**

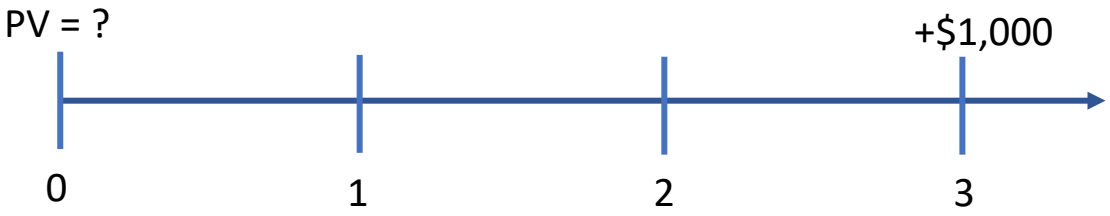
Make sure that your calculator is in BGN mode:
CF are expected at the beginning of each period: 2nd BGN, 2nd SET, 2nd QUIT

$$\text{FV}(1) = \mathbf{((1{,}000)(1.10))} + 1{,}000 = \$2{,}100$$
$$\text{FV}(2) = \mathbf{(1{,}000)(1 + 0.10)^2} + \mathbf{((1{,}000)(1.10))} + \mathbf{1{,}000} = \mathbf{\$3{,}310}$$
$$\text{FV}(3) = \mathbf{(1{,}000)(1 + 0.10)^3} + \mathbf{(1{,}000)(1 + 0.10)^2} + \mathbf{(1{,}000)(1.10)} = \mathbf{\$3{,}641}$$

Present value – LUMP SUM

Present value represents the value in present dollars of future cash flows discounted at an appropriate rate of return (the **discount rate**). The appropriate discount rate is the rate of return that the future cash flows could earn if they could be invested today.

Example 6. You are considering an investment that will pay $1,000 in 3 years from now. Similar investments (**same risk**) would generate a 10% annual return. What is the maximum you would be willing to pay today to invest in this project?



END	N	i/Y	PV	PMT	FV
Input	3	10		0	1,000
CPT			**-751.3**		

To answer this question, you must discount future $1,000 back to time zero.

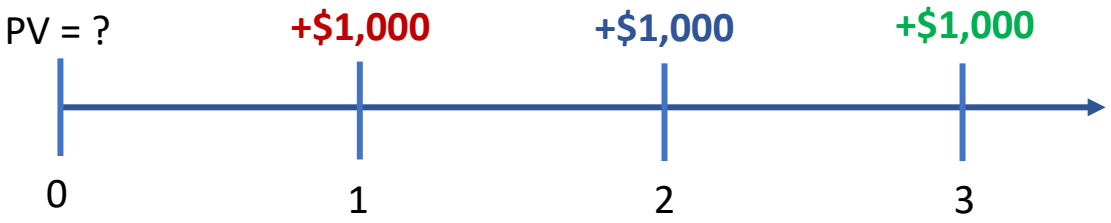
$$PV_0 = \frac{FV_N}{(1+I)^N}$$

$$PV_0 = \frac{1{,}000}{(1+0.10)^3} = 751.31$$

Conclusion: If you invest $751.31 today in this project, and receive $1,000 in 3 years, your annual rate of return will be 10%.

Present value – ORDINARY ANNUITY

Example 7. You are considering an investment that will pay $1,000 per year, **starting at the end of this year**, for the next 3 years. Similar investments (same risk) would generate a 10% annual return. What is the maximum you would be willing to pay today to invest in this project?



To answer this question, you must discount each future payments back to time 0.

PV of the 1st expected payment = 1,000 / (1.10) = 751.31

PV of the 2d expected payment = 1,000 / $(1.10)^2$ = 826.45

PV of the 3rd expected payment = 1,000 / $(1.10)^3$ = 909.09

Total PV = 2,486.85

END	N	i/Y	PV	PMT	FV
Input	3	10		1,000	0
CPT			-2,487		

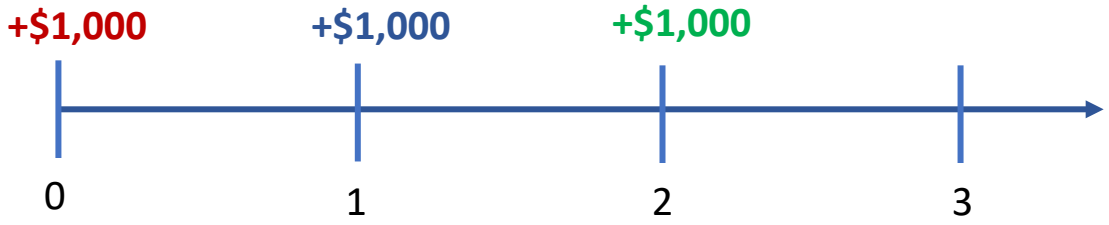
$$\text{PVA}_{\text{ordinary}} = \text{PVA} = \frac{\text{PMT}}{\text{I}}\left[1 - (1+\text{I})^{-\text{N}}\right]$$

$$\mathbf{PVA} = \frac{\mathbf{1,000}}{\mathbf{0.10}}\left[\mathbf{1 - (1 + 0.10)^{-3}}\right] = \mathbf{2,487}$$

Conclusion: If you invest $2,487 today in this project and receive $1,000 per year over the next 3 years, your annual rate of return will be 10%.

Present value – ANNUITY DUE

Example 8. You are considering an investment that will pay $1,000 per year, **starting today**, for the next 3 years. Similar investments (same risk) would generate a 10% annual return. What is the maximum you would be willing to pay today to invest in this project?



PV of the 1st expected payment = 1,000

PV of the 2d expected payment = 1,000 / (1.10) = 909.1

PV of the 3rd expected payment = 1,000 / $(1.10)^2$ = 826.4

Total PV = 2,736

BGN	N	i/Y	PV	PMT	FV
Input	3	10		1,000	0
CPT			**-2,736**		

$$\mathrm{PVA}^{\mathrm{due}} = \mathrm{PVA}_{\mathrm{ordinary}}(1+\mathrm{I})$$
$$= \frac{\mathrm{PMT}}{\mathrm{I}}\left[1-(1+\mathrm{I})^{-\mathrm{N}}\right](1+\mathrm{I})$$

$$\boldsymbol{PVA^{due}} = \frac{\mathbf{1,000}}{\mathbf{0.10}}\left[\mathbf{1-(1+0.10)^{-3}}\right]\mathbf{(1.10)}$$
$$\boldsymbol{= 2,735.54}$$

Conclusion: If you invest $2,487 today in this project and receive $1,000 per year over the next 3 years, your annual rate of return will be 10%.

Present value – constant perpetuity

Example 9. What is the present value of $1,000 per year received at the end of each year into infinity if the discount rate is 10%?

$$\text{PV}_{\text{perpetuity}} = \frac{\text{PMT}}{\text{I}}$$

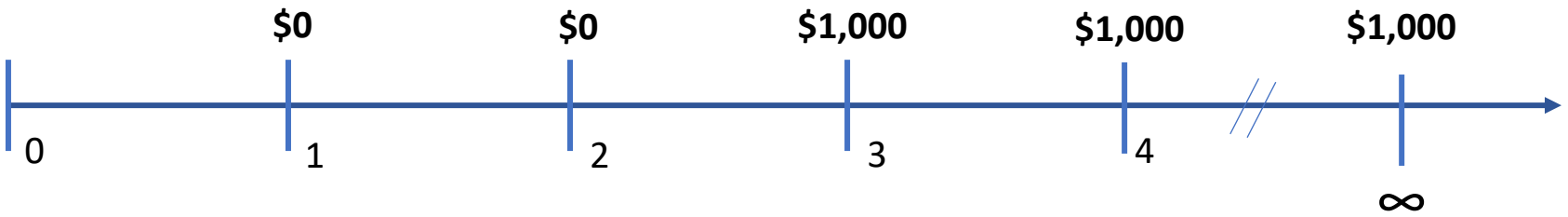
$$\mathbf{PV_{perpetuity}} = \frac{\mathbf{1,000}}{\mathbf{0.10}} = \mathbf{\$10,000}$$

END	N	i/Y	PV	PMT	FV
Input	500	10		1,000	0
CPT			-10,000		



Present value – delayed perpetuity

Example 10. Let's revisit example 9 and assume that the perpetuity start at time 3. What is the present value of the stream of expected cash flows if the discount rate is 10%?



PV_2 = 1,000/0.10

= 10,000

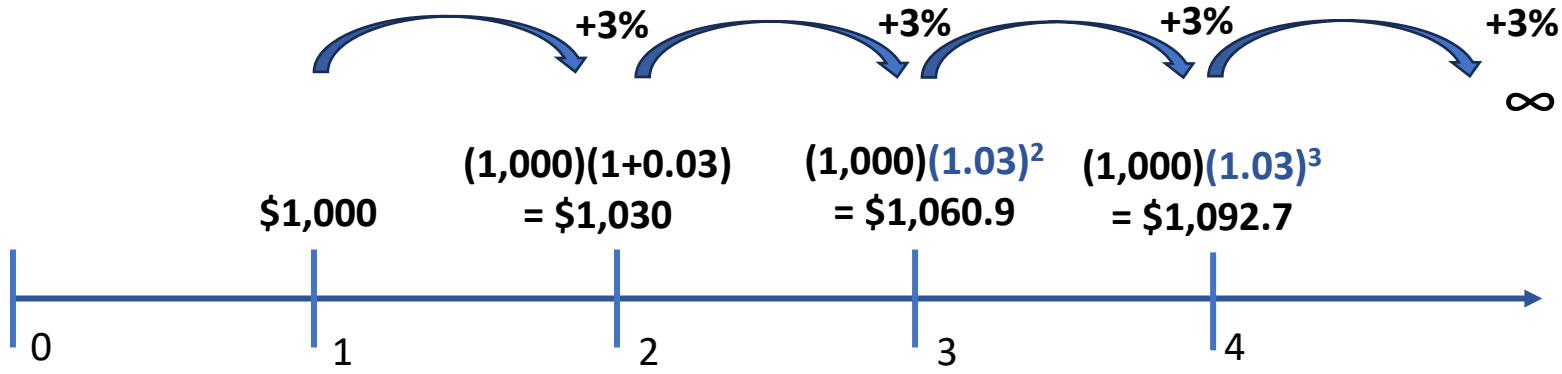
PV_0 = 10,000 / $(1.10)^2$

= 8,265

END	N	i/Y	PV	PMT	FV
Input	2	10		0	10,000
CPT			-8,265		

Present value – growing perpetuity

Example 11. Let's revisit example 9 and assume that the perpetuity payments will grow by 3% per year . What is the present value of the stream of expected cash flows if the discount rate is 10%?



PV_1 = CF2 / (I –g)

= 1,030 / (0.10 – 0.03)

= 14,714.3

PV_0 = 14,714.3/1.10

= $13,376.6

1st CF which is g percent higher than the previous one Is the 1st CF of the growing perpetuity.

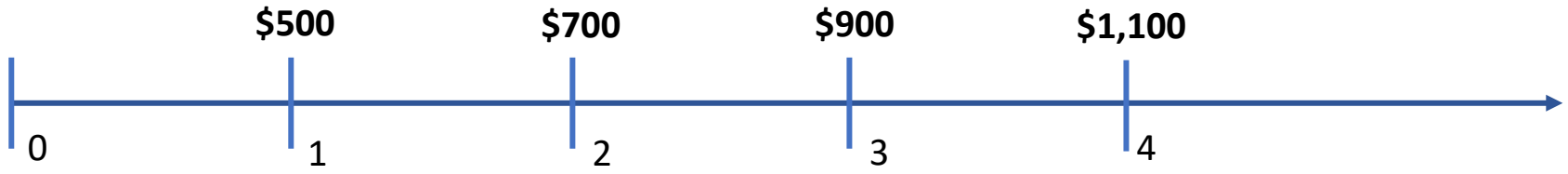
$$PV_{\text{at time zero}} = \frac{\text{PMT}_{\text{at time 1}}}{\text{I} - \text{g}}$$

$$PV_{\text{at time N}} = \frac{PMT_{\text{at time N+1}}}{I - g}$$

Present value and future value of uneven cash flows (1 of 3)

Example 12. Determine the **present value today** of the following **end of year**, **irregular** (**uneven**), **expected cash flows** if the proper discount rate is 10%:

Year 1	*$500*	*Year 3*	*$900*
Year 2	*$700*	*Year 4*	*$1,100*



Calculator keystrokes - Using the "CF" keys.

CF **CF0 = 0**
C01 = 500 **F01 = 1**
C02 = 700 **F02 = 1**
C03 = 900 **F03 = 1**
C04 = 1,100 **F04 = 1**

F stands for frequency.

NPV **enter I = 10** **CPT NPV= 2,460.56**

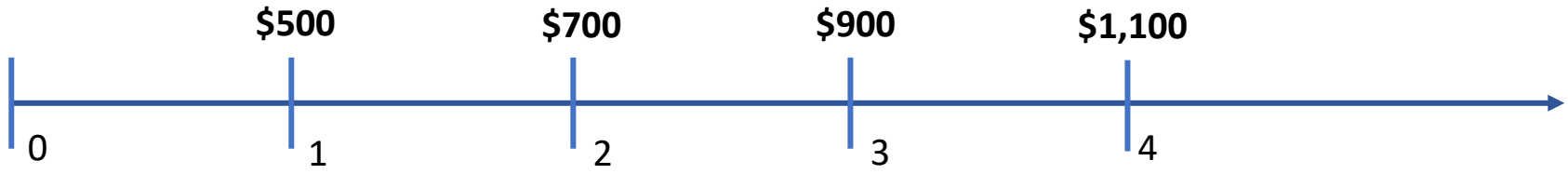
Use the "down arrow" to move to next register and press "Enter" when inputting values.

Note that all cash flows were input as positive values and that, unlike with the annuity keys, the NPV returned is also positive.

Present value and future value of uneven cash flows (2 of 3)

Example 12 end. Determine the **future value in 4 years** of the following **end of year**, **irregular** (**uneven**), **expected cash flows** if the proper discount rate is 10%:

Year 1	*$500*	*Year 3*	*$900*
Year 2	*$700*	*Year 4*	*$1,100*

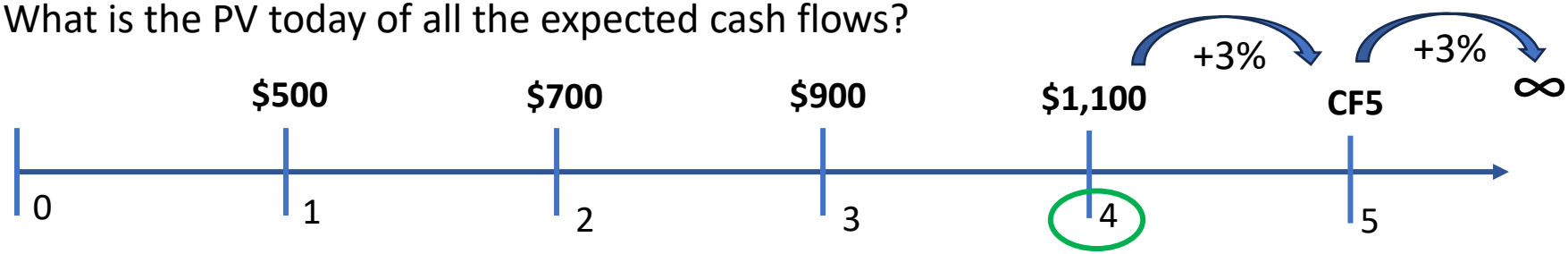


Step 1 – see previous slide: instead of computing the FV of each expected CF and calculate the total FV at time 3, we can use our previous calculation: **PV = 2,460.56**
Step 2 – convert this PV into dollars of time 3 (compute FV at time 3).

END	N	i/Y	PV	PMT	FV
Input	4	10	-2,460.56	0	
CPT					**3,602.5**

Present value and future value of uneven cash flows (3 of 3)

Example 13. Based on the information from the previous example, assume that after year four, the cash flows will grow at a constant rate of 3% (g) per year into infinity. What is the PV today of all the expected cash flows?



Step 1. Compute **PV at time 4** of the **growing perpetuity**.

$$PV_4 = \frac{CF_5}{r-g} = \frac{(1{,}100)(1+0.03)}{0.10-0.03} = \frac{1{,}133}{0.07} = 16{,}185.7$$

← PV at time 4 of CF5, 6, 7 …

Step 2. Use the **CF function** of your calculator to compute the total PV.

CF CF0 = 0

C01 = 500 F01 = 1

C02 = 700 F02 = 1

C03 = 900 F03 = 1

C04 = CF4 + PV_4

= 1,100 + 16,185.7 = 17,285.7 F04 = 1

NPV I = 10

CPT NPV

PV_0 = $13,515.6

Solving for the interest rate, the payment or the number of periods (1 of 3)

All time value of money calculations use at least four of the five following variables: PV, FV, I, N and PMT. If we have enough information, we can solve for any of the missing variables.

Example 14. You just invested $3,000 in a security that will yield 8% per year. Assuming that your annual returns are reinvested, how long will it take to double your investment?

END	N	i/Y	PV	PMT	FV
Input		8	-3,000	0	6,000
CPT	**9**				

OR, using the natural logarithm function (ln):

$$\$3,000(2) = (3,000)(((1+0.08)^t)$$

$$Ln(2) = (t)(Ln(1.08))$$

t = Ln(2)/Ln(0.08) = 0.6931/0.077 = 9

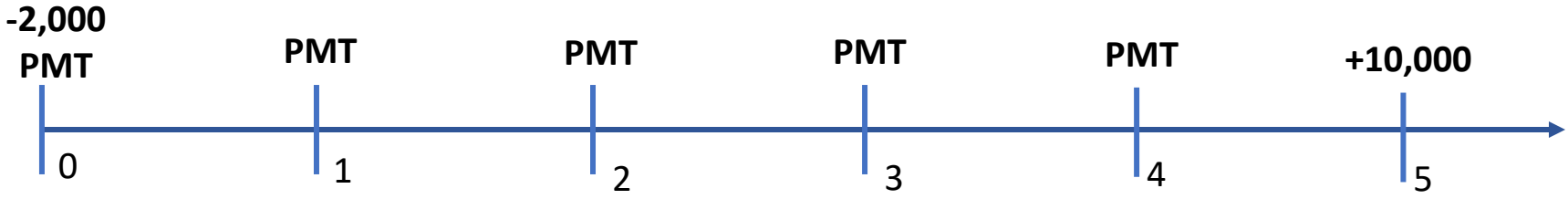
Solving for the interest rate, the payment or the number of periods (2 of 3)

Example 15. You just sold a stock investment for $50 per share. You paid $35 per share for this investment four years ago and did not receive any dividends during the period that you held the stock. What was your compound annual rate of return on this investment?

END	N	i/Y	PV	PMT	FV
Input	4		-35	0	50
CPT		9.33			

Solving for the interest rate, the payment or the number of periods (3 of 3)

Example 16. You have **$2,000** today (PV) in a savings account which pays **6%** per year. You would like to take 6 month trip to Europe in **5 years**. You think that you will need **$10,000** (FV) to finance this vacation. To reach your goal, you will add the same amount each year, always at the end of the year, to your savings account: 5 equal deposits(PMT) **starting today**. How large must each deposit be ?



BGN	N	i/Y	PV	PMT	FV
Input	5	6	-2,000		+10,000
CPT				**-1,225.6**	

Effective annual rate (EAR) and compounding periodicity (1 of 4)

Terminology

Nominal or Quoted Rate ($\mathbf{I_{NOM}}$): the rate quoted by a bank, broker, etc. Also called the **Annual Percentage Rate ("APR")**. It does not reflect the impact of any compounding and **is not used in calculations unless compounding occurs only once per year.**

Periodic Rate ($\mathbf{I_{PER}}$): the rate charged or earned each "period" e.g. each year, quarter, month, etc. The periodic rate is equal to the nominal rate divided by the number of compounding periods ("**M**") per year:

$$I_{PER} = \frac{I_{NOM}}{M}$$

Effective annual rate (EAR) and compounding periodicity (2 of 4)

Effective Annual Rate ("EAR" or "EFF"): the annual rate that takes into account any compounding. EAR allows us the compare APRs that have different compounding periods. The formula is:

$$\text{EAR} = \left[1 + \frac{\text{I}_{\text{NOM}}}{\text{M}}\right]^{\text{M}} - 1$$

Where:

$\text{M} = \text{Number of compounding periods per year}$

The EAR adjusts the APR for the number of compounding periods per year and expresses the interest rate as if it were compounded once per year. Another way to think about the EAR is that it is the rate that converts a present value into a future value over a one-year period.

Effective annual rate (EAR) and compounding periodicity (3 of 4)

If there is only one compounding period per year, the APR and the EAR are the same.

As the number of compounding periods increase, the EAR starts to rise relative to the nominal rate.

For example, here are the EARs at different compounding intervals for a quoted rate of 8%

Compounding interval	Effective annual rate
Annual	8.0000%
Semi-annual (2 times per year)	8.1600%
Quarterly	8.2432%
Monthly	8.3000%
Daily	8.3278%
Continuous*	8.3287%

Continuous compounding assumes that interest is being credited the instant it is earned and therefore, the amount of interest grows continuously. Continuous compounding represents the upper limit for compounding intervals.

EAR with continuous compounding = $e^{\text{APR}} - 1$.

Effective annual rate (EAR) and compounding periodicity (4 of 4)

Example 17. What is the EAR of 12% compounded monthly?

$$\mathbf{EAR} = \left(1 + \frac{0.12}{12}\right)^{12} - 1 = 0.126825\ (12.6825\%)$$

Using your financial calculator

2nd ICONV

NOM = 12 = quoted rate

C/Y = 12 = compounding periodicity = 12 times per year

CPT EFF = 12.6825

Loan amortization (1 of 6)

Loans that are repaid in fixed payments are called **amortized loans**. The payments are **blended**: they cover both interest payment and principal repayment. At the end, loan has been fully repaid.

The **Texas Instruments BA II Plus has a loan amortization function** that allows us to compute the loan payment as well as the following:

- The balance of the loan after any payment.
- The interest and principal amounts of any payment.
- The amount of principal and interest paid over a range of payments.

Terminology – non amortized loans

Interest-only loan (Bullet loan): the borrower pays only interest throughout the term, then repays the entire principal in one lump sum ("bullet") at maturity.
Balloon loan: regular payments are smaller than for a fully amortizing loan, leaving a large final "balloon" payment of remaining principal at the end.

Loan amortization (2 of 6)

If the interest compounding interval corresponds to the payment interval (e.g. monthly compounding and monthly payments), the calculations are fairly straightforward.

If the **compounding intervals are different than the payment intervals**, we must use the P/Y and C/Y functions to get the correct answer.

A difference between payment frequency and compounding frequency occurs most often with **mortgage loans**. **By law, the interest rate on Canadian fixed rate mortgages must be compounded semi-annually.** However, **most mortgage payments are monthly**. The following example illustrates how we deal with this complexity

Loan amortization (3 of 6)

Example 18. Determine the monthly payment for a $200,000 fixed rate mortgage with a quoted annual rate of 6%, compounded semi-annually, and a 25-year amortization period.

P/Y = 12 C/Y = 2	N	i/Y	PV	PMT	FV
Input	25(12) = 300	6	200,000		0
CPT				**-1,279.6**	

OR

P/Y = 1 C/Y = 1	N	i/Y	PV	PMT	FV
Input	300	0.49386	200,000		0
CPT				**-1,279.6**	

$$Effective\ monthly\ rate = \left(1 + \frac{0.06}{2}\right)^{2/12} - 1 = 0.49386\%$$

Loan amortization (4 of 6)

Example 18 continued. The loan is an amortized loan. E**arly payments** contain **mostly interest** and a **small amount of principal repayment**. As the principal is progressively repaid, the interest portion of the payment decreases, and the portion of the payment applied to the loan principal repayment increases. See below: loan amortization schedule prepare in Excel and available on the Learning hub.

Payment number	Beginning Balance	Blended Payment	Interest	Principal	Ending Balance
0					$ 200,000.00
1	$200,000.00	-1,279.61	-987.72	-291.89	$ 199,708.11
2	$199,708.11	-1,279.61	-986.28	-293.33	$ 199,414.78
3	$199,414.78	-1,279.61	-984.83	-294.78	$ 199,120.00
4	$199,120.00	-1,279.61	-983.38	-296.23	$ 198,823.77
5	$198,823.77	-1,279.61	-981.92	-297.70	$ 198,526.07
6	$198,526.07	-1,279.61	-980.45	-299.17	$ 198,226.90
7	$198,226.90	-1,279.61	-978.97	-300.65	$ 197,926.26
8	$197,926.26	-1,279.61	-977.48	-302.13	$ 197,624.13
9	$197,624.13	-1,279.61	-975.99	-303.62	$ 197,320.50
10	$197,320.50	-1,279.61	-974.49	-305.12	$ 197,015.38
11	$197,015.38	-1,279.61	-972.98	-306.63	$ 196,708.75
12	$196,708.75	-1,279.61	-971.47	-308.14	$ 196,400.61

Loan amortization (5 of 6)

Example 18 continued. The loan is an amortized loan. E**arly payments** contain **mostly interest** and a **small amount of principal repayment**. As the principal is progressively repaid, the interest portion of the payment decreases, and the portion of the payment applied to the loan principal repayment increases. See below: loan amortization schedule prepare in Excel and available on the Learning hub.

Payment number	Beginning Balance	Blended Payment	Interest	Principal	Ending Balance
288	$ 16,073.82	-1,279.61	-79.38	-1,200.23	$ 14,873.59
289	$ 14,873.59	-1,279.61	-73.46	-1,206.16	$ 13,667.43
290	$ 13,667.43	-1,279.61	-67.50	-1,212.11	$ 12,455.32
291	$ 12,455.32	-1,279.61	-61.51	-1,218.10	$ 11,237.21
292	$ 11,237.21	-1,279.61	-55.50	-1,224.12	$ 10,013.10
293	$ 10,013.10	-1,279.61	-49.45	-1,230.16	$ 8,782.94
294	$ 8,782.94	-1,279.61	-43.38	-1,236.24	$ 7,546.70
295	$ 7,546.70	-1,279.61	-37.27	-1,242.34	$ 6,304.35
296	$ 6,304.35	-1,279.61	-31.13	-1,248.48	$ 5,055.88
297	$ 5,055.88	-1,279.61	-24.97	-1,254.64	$ 3,801.23
298	$ 3,801.23	-1,279.61	-18.77	-1,260.84	$ 2,540.39
299	$ 2,540.39	-1,279.61	-12.55	-1,267.07	$ 1,273.32
300	$ 1,273.32	-1,279.61	-6.29	-1,273.32	$ 0.00

Loan amortization (6 of 6)

Example 18. end Determine the monthly payment for a $200,000 fixed rate mortgage with a quoted annual rate of 6%, compounded semi-annually, and a 25-year amortization period.

What will the **balance** of this loan be **after the first year of payments**?

2nd AMORT P1 = 1 P2 = 12 BAL= 196,400.61

How much of the **15th payment** will be applied to the principal of the loan?

2nd AMORT P1 = 15 P2 = 15 PRN = -312.73

What amounts of **principal and interest** will be **paid during the third year** of payments?

2nd AMORT P1 = 25 P2 = 36 PRN = -4,051.15

INT = -11,304.21

Use of P/Y and C/Y registers: Make sure to reset them to "1" after every problem or you are likely to get the wrong answer on subsequent problems. This is important when writing an exam or quiz.

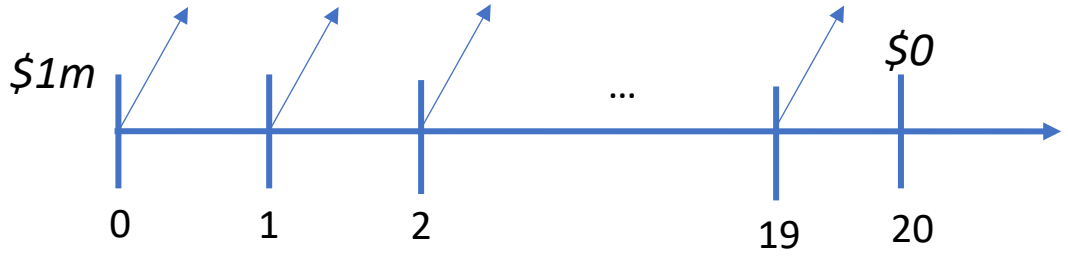
Financial planning and growing annuities (1 of 3)

A **growing annuity** can be a very useful concept in personal financial planning.

It is often used to factor in the **impact of expected inflation** to calculate the **real** or **inflation-adjusted** income available from a portfolio or investment fund.

Example 19. You are 65 years old and have saved **$1,000,000** for your retirement. You expect to live another **20 years**, earn an **8% annual rate** on your portfolio and believe that **inflation** will average **3%** per year over the next 20 years. You want to receive a **constant inflation-adjusted income** each year. How much should your initial withdrawal amount be, made today, so that you do not run out of money?

Amount withdrawn each year, at the beginning of the year, has the same purchasing power

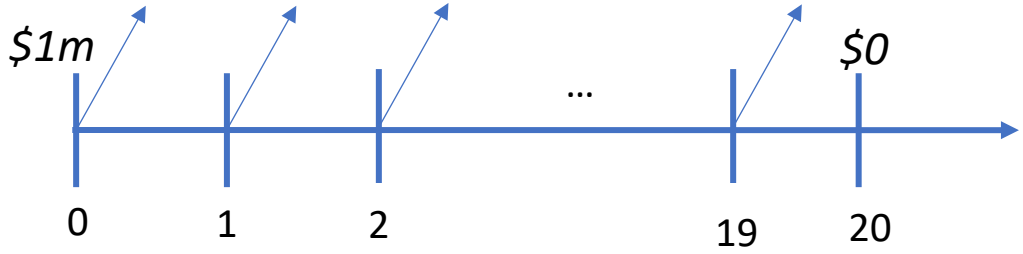


For example:
if you withdraw $100 at time zero,
you'll withdraw 100(1.03) = $103 at time 1,
You'll withdraw 100(1.03)(1.03) = $106.09 at time 2 …

Financial planning and growing annuities (2 of 3)

Example 19 continued. You are 65 years old and have saved **$1,000,000** for your retirement. You expect to live another **20 years**, earn an **8% annual rate** on your portfolio and believe that **inflation** will average **3%** per year over the next 20 years. You want to receive a **constant inflation-adjusted income** each year. How much should your initial withdrawal amount be, made today, so that you do not run out of money?

Amount withdrawn each year, at the beginning of the year, has the same purchasing power



Step 1. Find the inflation-adjusted rate of return (the "real" rate of return)

$$1 + r_{NOM} = (1 + \textit{Real rate})(1 + \textit{expected inflation})$$

$$\textbf{Real rate} = \left(\frac{1 + \text{r}_{\text{NOM}}}{1 + \text{expected inflation}}\right) - 1 = \frac{1.08}{1.03} - 1 = 0.0485437\ (4.85437\%)$$

Financial planning and growing annuities (3 of 3)

Example 19 end. You are 65 years old and have saved **$1,000,000** for your retirement. You expect to live another **20 years**, earn an **8% annual rate** on your portfolio and believe that **inflation** will average **3%** per year over the next 20 years. You want to receive a **constant inflation-adjusted income** each year. How much should your initial withdrawal amount be, made today, so that you do not run out of money?

BGN	N	i/Y	PV	PMT	FV
Input	20	4.85437	-1,000,000		0
CPT				**-75,585.54**	

